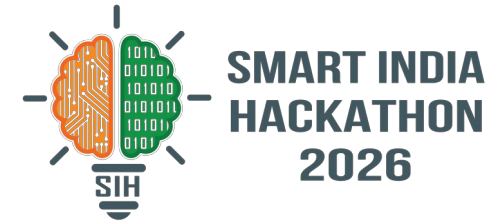
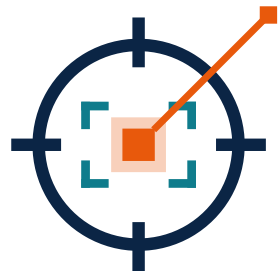


SMART INDIA HACKATHON 2026



- **Problem Statement ID** – 26169
- **Problem Statement Title** – Development of an AI-Based Virtual Camera Tracking System for Coarse Alignment of Mobile Free Space Optical Communication (FSOC) Terminals
- **Theme** – Smart Automation, Space Technology
- **PS Category** – Software
- **Team ID** – [fill in]
- **Team Name** – Blank Point



LAKSHYA

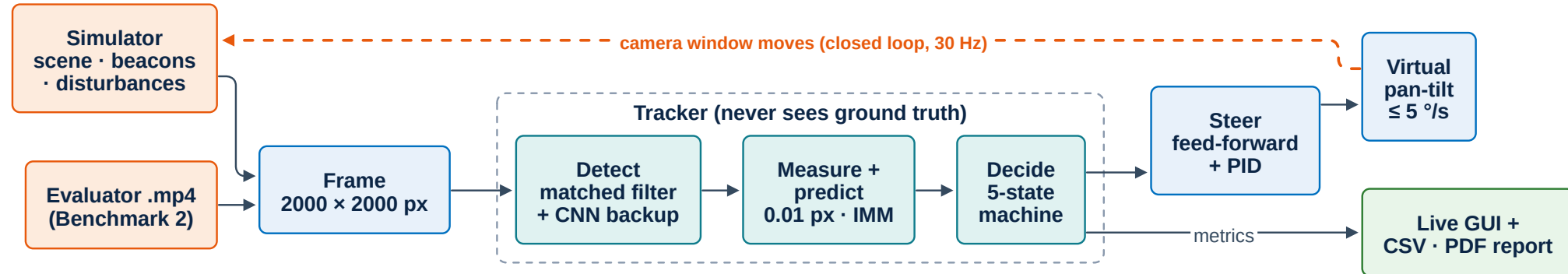
AI virtual camera tracker for coarse alignment of mobile FSOC terminals



LAKSHYA: AI-Assisted Virtual Camera Tracker

Find the beacon, measure it to 0.01 px, predict it, and steer a 5 °/s virtual pan-tilt camera to keep it centred

- **Proposed Solution** A standalone desktop app (plus the same program as a web app) that **renders a 2000 × 2000 px sky with a moving beacon and real-world disturbances**, then detects, measures, predicts and steers a rate-limited 640 × 480 virtual camera to keep that beacon centred. Every run writes its own performance report. Evaluator .mp4 files plug into the same pipeline.



- **How it addresses the problem** *the PS's four coarse-alignment steps*

OBSERVE

Tracker watches the whole 2000 × 2000 px (12.5°) scene each frame, as the PS asks ("observe the environment"); hard mode limits it to the window.

ACQUIRE

Matched filter + noise-adaptive threshold; a spot must hold 3 of 4 frames before lock. Locks in 0.7 to 1.4 s (spec ≤ 2 s).

ESTIMATE

Centre to ≈ 0.01 px by 2D Gaussian fit; an IMM filter predicts line, circle, figure-8 and random paths unaided.

ADJUST

Feed-forward + PID leads the beacon by the gimbal's 1-frame delay, never above 5 °/s: 6.4 to 8.2 px error (spec ≤ 10 px).

- **Innovation and uniqueness**

Physics-ordered disturbances

Atmosphere, then platform sway and jitter, then sensor noise: the order real light meets them.

Self-trained AI safety net

84 k-parameter CNN trained on our own simulator frames with exact labels; no dataset needed.

Provably blind to the truth

Ground truth deleted before tracking gives identical results to 10⁻⁶ px.

Track-before-detect

Links 3 to 6 σ spots across frames. Faint-beacon lock went from never to 91–97.5 %.

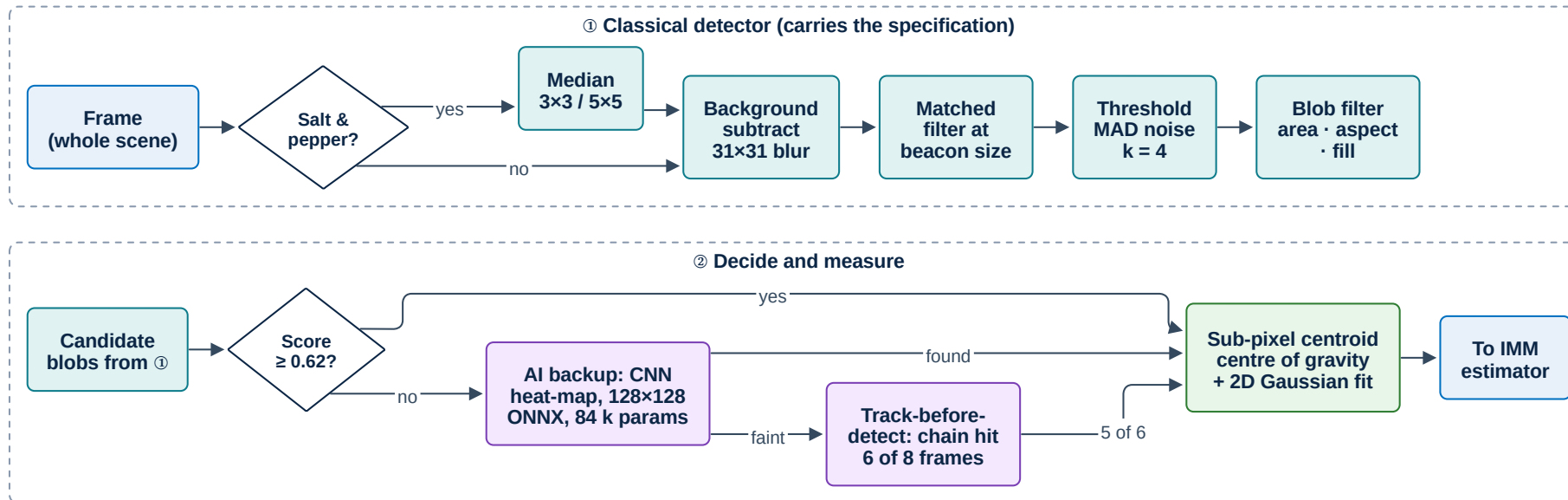
One pipeline, two sources

An evaluator .mp4 replaces the simulator; the tracker code runs unchanged (Benchmark 2).

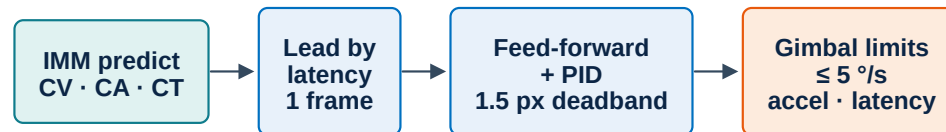
Holds identity among decoys

Appearance signature + a whole-scene check every 0.5 s keeps the right beacon.

Every frame: detect and measure *classical vision carries the spec; AI fills the gaps*



Then estimate and steer *IMM prediction, lead, command, physical limits*

**22.5"**

one pixel (4° / 640 px)

26.7 px

max slew per frame at 5 °/s, 30 Hz

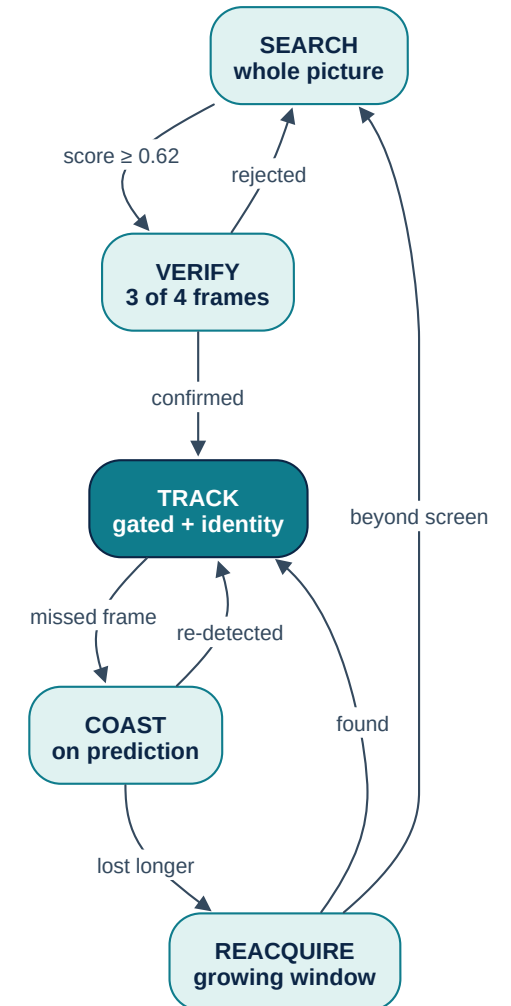
0.95 s

centre to corner, target known

50 ms

per frame at 20 FPS; we need 4 to 12

Tracker states



■ Software architecture *modular Python packages, one code path for GUI, CLI and tests*

DISPLAY · desktop and web built from one interface definition

Desktop app
PyQt6 + pyqtgraph

Web app
FastAPI + Uvicorn

Command line
run · video · batch

PyInstaller builds
Win · Linux · macOS x2

INSTRUMENTATION · log per run: duration, FPS, acquisition, re-acquisition, mean / max / RMSE error, lock, processing time

telemetry
per-frame record

metrics
pass / fail vs PS

report
auto PDF + CSV

naming
results/<run> folder

CONTROL + PERCEPTION · the tracker, blind to ground truth

detect
classical + CNN

estimator
IMM CV·CA·CT

tracker
5 states · identity

controller
FF + PID

WORLD + SOURCES · FrameSource: simulator or .mp4

scene
+ 8 paths

camera
gimbal limits

disturbance
+ renderer

sources
video reader

ENGINE · one typed config field per PS row (all 25 rows) · seeded, repeatable loop

config
YAML scenarios

simulation
fixed step

scenario pack
13 YAML + template

tests
22 + regression batch

■ Technologies used

LANGUAGE + CORE

Python 3.12

NumPy

SciPy

PyYAML

COMPUTER VISION

OpenCV

sub-pixel Gaussian fit

AI

PyTorch (train only)

ONNX

ONNX Runtime

DESKTOP GUI

PyQt6

pyqtgraph

WEB APP

FastAPI

Uvicorn

Caddy

REPORTS

Matplotlib PDF

CSV

JSON

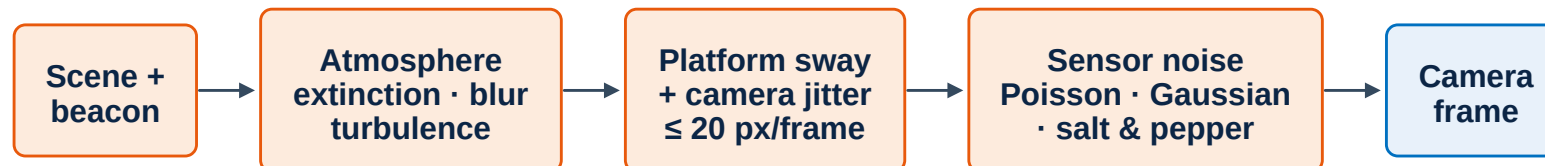
BUILD + QA

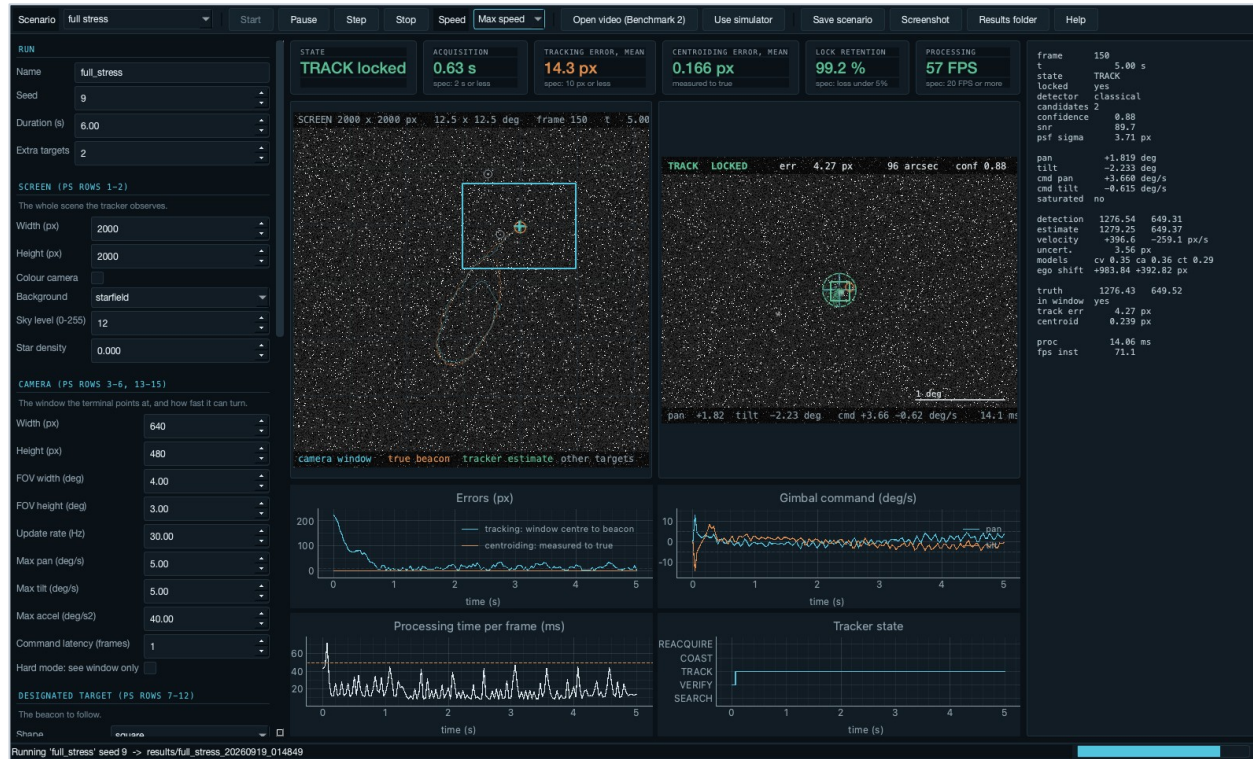
PyInstaller

pytest

GitHub Actions

■ Disturbances applied in physical order *PS rows 21 to 25, each switchable*





Desktop app mid-run, multi-target stress scenario (haze, 3 noise types, sway, jitter, 2 decoys): TRACK locked, 0.63 s acquisition, 99.2 % lock, 57 FPS with live drawing; 14.3 px error is the injected shake

Measured on clear line, circle and figure-8 runs (15 s, 3 seeds each); noise, fog and low-light runs stay inside every spec

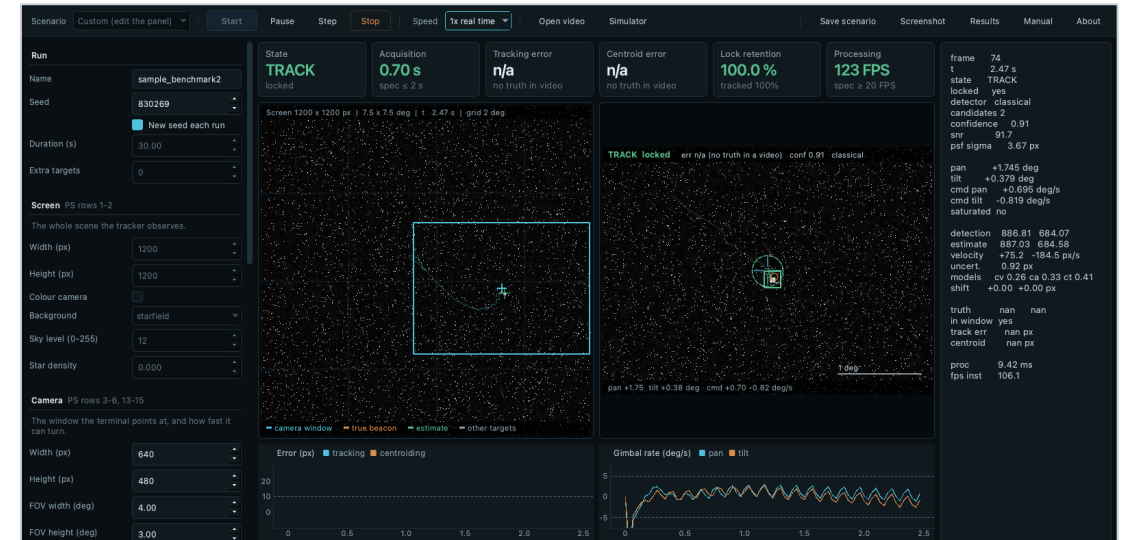
0.7–1.4 s
acquisition
spec ≤ 2 s

6.4–8.2 px
tracking error
spec ≤ 10 px

100 %
lock retention
spec: loss < 5 %

0.006 px
centroid error, clear
0.1–0.4 px heavy noise

80–250
FPS on a laptop
spec ≥ 20 FPS



Benchmark 2 input: sample_benchmark2.mp4 (1200 × 1200 px, 30 fps) opened directly. The simulator is bypassed; size, frame rate and rotation come from the file. Our measured beacon centre is logged every frame (det_x, det_y) to compare with the predefined values.

Source code: github.com/tanmayhutt/SIH26169 (private; access on request)

Web app: sih26169.blankpoint.club (login)

Demo video (≈ 4 min): sih26169.blankpoint.club/downloads (login)

FEASIBILITY AND VIABILITY

■ Feasibility: already built and measured *15 s runs, 3 seeds, laptop CPU*

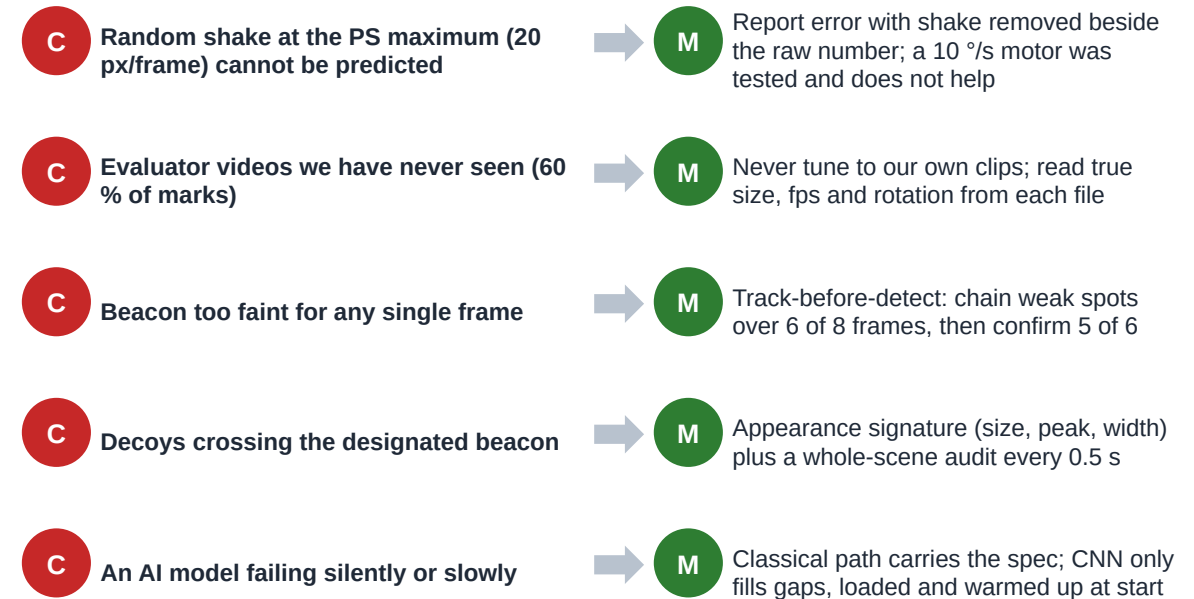
Scenario	Acquire	Error	Lock	Verdict
Clear: line, circle, figure 8	0.7–1.4 s	6.4–8.2 px	100 %	Pass
Clear random walk	≈ 1.1 s	≈ 10 px	98 %	At limit
Heavy noise (S&P 10 %, σ 20, Poisson)	0.8–1.1 s	7.3–8.2 px	100 %	Pass
Fog and low light	0.7–1.7 s	6.6–8.0 px	100 %	Pass
Faint beacon, 10 seeds	1.3–5.5 s	6.7–12.3 px	91–97.5 %	Most seeds
Decoys + haze + noise + sway	0.7–1.3 s	13.6–14.4 px	98–99 %	Identity held
Platform at PS max (20 + 20 px/frame)	≈ 0.8 s	≈ 25 px	76–88 %	Physical limit
Hard mode (camera window only)	3–12 s	7–12 px	100 %	By design

Viability: runs offline on an ordinary laptop CPU, no camera, pan-tilt unit or optics to buy. Same numbers on Windows, Linux and both Macs; every build ships only after its own packaged smoke test.

■ How every change is verified before it is accepted



■ Challenges, risks and our strategy



■ Potential impact on the target audience

ISRO SAC and DoS PAT teams

Develop and compare coarse-pointing algorithms before any camera or gimbal is bought.

Universities and students

A free, repeatable PAT lab: the PS's own "algorithm development and learning" goal.

Indian FSOC, UAV and NewSpace start-ups

Benchmark trackers on the same scenario pack and per-frame logs.

Defence and disaster communications

Rapid optical links between vehicles and drones need a fast coarse lock.

■ Benefits of the solution

Economic

Replaces a camera, pan-tilt and optics bench for algorithm work with one laptop.

Technical

Seeded, repeatable runs; per-frame CSV logs make trackers directly comparable.

Social

Runs offline on Windows, Linux and macOS with nothing to install.

Environmental

Fewer hardware prototypes; optical links use licence-free spectrum.

Strategic

Indigenous PAT software for India's optical and quantum link work [14].

Ready for hardware

FrameSource and gimbal are interfaces: a real camera and PTU plug in.

■ Why coarse pointing matters: optical links are already here

200 Gbps

NASA TBIRD space-to-ground laser link, 2023 [11]

20 Gbps · 20 km

Taara free-space optical links in Airtel's network in India [15]

1.8 Gbit/s

ESA EDRS laser relay, locking onto satellites 35,000+ km away [13]

■ Path from virtual bench to a real link



■ FSOC and pointing, acquisition and tracking

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- [2] M. A. Khalighi and M. Uysal, "Survey on free space optical communication: A communication theory perspective," IEEE Commun. Surveys Tuts., vol. 16, no. 4, pp. 2231–2258, 2014. [doi:10.1109/COMST.2014.2329501](https://doi.org/10.1109/COMST.2014.2329501)
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- [4] R. Abdelfatah, N. Alshaer, and T. Ismail, "A review on pointing, acquisition, and tracking approaches in UAV-based FSO communication systems," Opt. Quantum Electron., vol. 54, art. 571, 2022. [doi:10.1007/s11082-022-03968-2](https://doi.org/10.1007/s11082-022-03968-2)
- [5] L. C. Andrews and R. L. Phillips, Laser Beam Propagation through Random Media, 2nd ed. SPIE Press, 2005. [doi:10.1117/3.626196](https://doi.org/10.1117/3.626196)

■ Tracking, estimation and vision methods

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- [9] S. Thomas et al., "Comparison of centroid computation algorithms in a Shack-Hartmann sensor," Mon. Not. R. Astron. Soc., vol. 371, no. 1, pp. 323–336, 2006. [doi:10.1111/j.1365-2966.2006.10661.x](https://doi.org/10.1111/j.1365-2966.2006.10661.x)
- [10] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," MICCAI 2015, LNCS 9351, pp. 234–241. [doi:10.1007/978-3-319-24574-4_28](https://doi.org/10.1007/978-3-319-24574-4_28)

■ Missions and deployments

- [11] NASA, "NASA, partners achieve fastest space-to-ground laser comms link" (TBIRD, 200 Gbps), 2023. nasa.gov/.../fastest-space-to-ground-laser-comms-link
- [12] NASA, "NASA's laser comms demo makes deep space record, completes first phase" (DSOC), 2024. nasa.gov/.../dsoc
- [13] ESA, "Europe's SpaceDataHighway relays first Sentinel-1 images via laser" (EDRS), 2016. esa.int/.../EDRS
- [14] Dept. of Space / ISRO, "Entanglement based quantum communication over 300 m free space" (SAC and PRL), 2022. isro.gov.in/DeptofSpace.html
- [15] Taara, "Airtel taps into Taara's free-space optical communications," 2023. taaraconnect.com/post/airtel-taps-into-taara...

■ Software and project documents

- [16] ONNX Runtime; G. Bradski, "The OpenCV Library," Dr. Dobb's J. Softw. Tools, 2000. onnxruntime.ai
- [17] SIH 2026 Problem Statement 26169, ISRO Space Applications Centre (parameter table and evaluation method, rows 1 to 25).
- [18] LAKSHYA Technical Report (12 pp.), User Manual and per-run performance logs, in our repository.

■ Coverage, definitions and known limits

Coverage: all 25 PS table rows, 8 of 8 "shall" functions, all 5 deliverables and the optional video; extras include 7 paths + waypoints, 3 beacon shapes, colour camera and up to 8 decoys.

Definitions: lock = TRACK state within 30 px of the camera centre; tracking error = true beacon to camera centre; centroiding error = our measured centre to the true centre (the PS leaves these open).

Known limits: one faint-beacon seed in 10 acquires in 5.5 s; hard mode searches for 3 to 12 s; a faint beacon that never moves is not covered; the builds are unsigned, so the OS warns on first launch.